The University of Minnesota's Traumatic Stress Screen for Children and Adolescents (TSSCA):

A Preliminary Analysis

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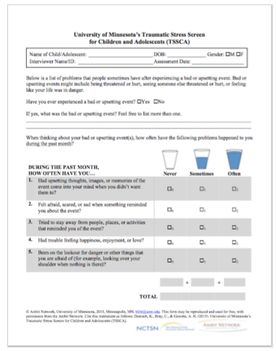
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Background

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Unrecognized and untreated childhood trauma can lead to the development of trauma-related symptomologies such as distressing recurrent memories and nightmares, physical reactions to trauma reminders, persistent negative emotions, concentration difficulties, and reckless behavior. These symptoms can persist and worsen over time. One way to identify trauma-exposed youth and provide necessary access to evidence-based mental health treatment is through systematic trauma screening in child-service systems, including schools, child welfare, and juvenile justice. The University of Minnesota's Traumatic Stress Screen for Children and Adolescents (TSSCA) was designed to provide practitioners with an efficient yet precise 5-item tool for screening and identifying children, ages 5 to 18 years, that may have experienced a traumatic event and are in need of services.

The TSSCA



University of Minnesota's Traumatic Stress Screen for Children and Adolescents (TSSCA)

Name of Child/Adolescent: _____ DOB: _____ Gender: ☐M ☐F

Interviewer Name/ID: _____ Assessment Date: _____

Below is a list of problems that people sometimes have after experiencing a bad or upsetting event. Bad or upsetting events might include being threatened or hurt, seeing someone else threatened or hurt, or feeling like your life was in danger.

Have you ever experienced a bad or upsetting event? ☐Yes ☐No

If yes, what was the bad or upsetting event? Feel free to list more than one.

When thinking about your bad or upsetting event(s), how often have the following problems happened to you during the past month?

DURING THE PAST MONTH, HOW OFTEN HAVE YOU …	Never	Sometimes	Often
1. Had upsetting thoughts, images, or memories of the event come into your mind when you didn't want them to?	☐0	☐1	☐2
2. Felt afraid, scared, or sad when something reminded you about the event?	☐0	☐1	☐2
3. Tried to stay away from people, places, or activities that reminded you of the event?	☐0	☐1	☐2
4. Had trouble feeling happiness, enjoyment, or love?	☐0	☐1	☐2
5. Been on the lookout for danger or other things that you are afraid of (for example, looking over your shoulder when nothing is there)?	☐0	☐1	☐2
	+	+	
TOTAL			

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Figure 1. The TSSCA form.

What is Screening?

According to NCTSN, "Trauma-informed screening refers to a brief, focused inquiry to determine whether an individual has experienced specific traumatic events or reactions to trauma" (2012). These screens are not diagnostic. Instead, the intention is to identify children who *may* be experiencing traumatic stress symptomology. Results are used as a basis for referral for a more thorough assessment conducted by a trained mental health professional or physician (Conradi, Wherry, & Kisiel, 2011, page 132). Following assessment, children who demonstrate need should be connected to appropriate services.

Need

Without trauma-informed screening, trauma exposure and symptomatology in youth may go undetected, leading front line service providers to treat symptoms and disorders that may only partially explain an individual's presentation. Such screening can also assist mental health and other front-line service providers to identify youth at risk of developing more pervasive and severe symptoms of traumatic stress (SAMHSA, 2014). In addition to identifying at-risk children, screening for traumatic stress can prevent misdiagnosis and inappropriate treatment planning (SAMHSA, 2014).

Methods

Development

After extensive review of a broad range of diagnostic tools, the research team used item response modeling to create a 15-item instrument with questions representing symptoms of traumatic stress. Multiple stepwise regression was used to extract the 5 top performing items from the 15-item screener.

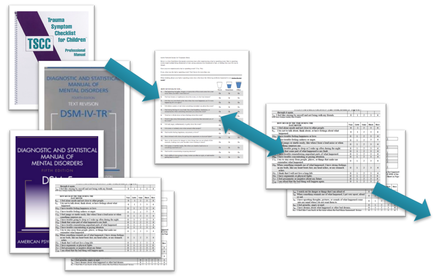


Figure 2. Diagram illustrating the development process of the TSSCA.

Sample

- 130 individuals seen in community mental health settings.
- 67 female, 53 male
- Predominantly white (n=83), non-Hispanic
- Representing exposures and symptoms related to a wide variety of trauma types (e.g., domestic violence, neglect, school violence, bereavement, etc.)

Measures

- University of California Post Traumatic Stress Disorder Reaction Index 5 (UCLA-5): a 31-item questionnaire that assesses for symptoms of PTSD in children and adolescents. The UCLA is considered the gold standard assessment for diagnosing PTSD in children and adolescents.
- Sociodemographic questionnaire: A clinician completed brief questionnaire about child age, gender, trauma type, and race/ethnicity.

Data Analysis

- To test for reliability in the TSSCA, coefficient Omega was estimated using R (Dunn, et al., 2013).
- To test for discriminative and criterion validity (i.e. predictive validity), a ROC (Receiver Operating Characteristic; Hsiao, Bartko, & Potter, 1989) Curve Analysis was performed using the SPSS package Version 22 (IBM Corp., Armonk, NY). The UCLA -5 was used as the "gold standard of diagnosis" to define true and false cases.
- AUCs for the 5-item TSSCA and the original 15-item version were compared to decide whether the shortened TSSCA version performs equivalently to the full-length version.
- There were no missing values in the variables used for analyses. 28 duplicate cases (e.g. follow-ups) were removed to ensure the assumption of independence.

Results

Reliability and Validity

- The value of Omega was .81, indicating an acceptable level of internal consistency within the TSSCA (Dunn, et al., 2013).
- The AUC for the TSSCA was .87 (SE = .04, 95% Confidence Interval= [.80, .95]) indicating very good inherent discriminative validity (Hajian-Tilaki, 2013).
- The AUC for the original 15-item version is .89 (SE = .03, 95% Confidence Interval= [.83, .95]). Visually, the curves for the 5-question and 15-question screens are similar in area (see figure 1). Further statistic testing indicated that the difference between those two AUCs was non-significant (Z statistic = .32, SE = .01, p = .75; DeLong, et al., 1988) suggesting that the shortened screener (i.e. the TSSCA) functioned equivalently to the full-length version in identifying youth in need of continued trauma evaluation.

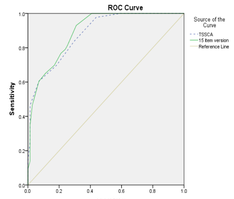


Figure 3. The ROC curves for the 5 item and 15 item versions of the TSSCA.

Determination of Optimal Cutoff

An optimal cutoff score, with no constraints on cost, benefit and prevalence rates, is the score at which the optimal balance between sensitivity and specificity was achieved, depending on the purpose of the screener. Several potential optimal cutoffs were identified for the TSSCA, including 5, 6 and 7. The authors determined that a score of 6 was the optimal cutoff for the TSSCA. With a cutoff of 6, the sensitivity of the of the screener was 82.76%, specificity was 85.42%, positive predictive value was 87.27% and negative predictive value was 80.39%.

Table 1.

The screener performance indices for potential cutoffs.

Potential Cutoffs	Sensitivity	Specificity	PPV	NPV	Youden J	d^2
1	100.00%	6.25%	56.31%	100.00%	0.06	0.88
2	100.00%	16.67%	59.18%	100.00%	0.17	0.69
3	100.00%	39.58%	66.67%	100.00%	0.40	0.37
4	96.55%	52.08%	70.89%	92.59%	0.49	0.23
5	89.66%	64.58%	75.36%	83.78%	0.54	0.14
6	82.76%	85.42%	87.27%	80.39%	0.68	0.05
7	67.24%	89.58%	88.64%	69.35%	0.57	0.12
8	27.59%	91.67%	80.00%	51.16%	0.19	0.53
9	20.69%	97.92%	92.31%	50.54%	0.19	0.63
10	20.69%	97.92%	92.31%	50.54%	0.19	0.63
11	NA	NA	NA	NA	NA	NA
12	NA	NA	NA	NA	NA	NA

Note. PPV stands for Positive Predictive Value, NPV stands for Negative Predictive Value. Values for cutoffs of 11 and 12 are not available due to the empty cells in crosstabulation.

References and additional resources available upon request from the author.
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